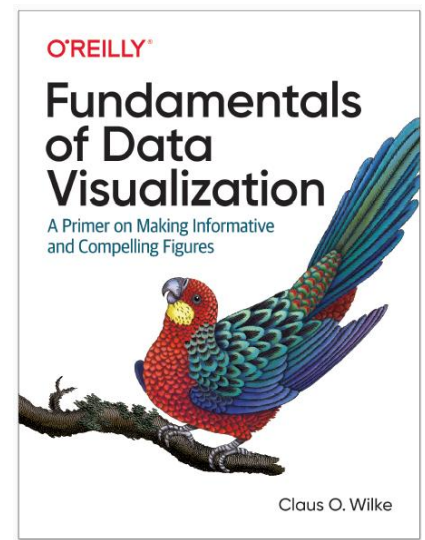


Introduction to Data Visualization

Veera Muangsin

References

- Book: Fundamentals of Data Visualization, Claus O. Wilke, 2019.
clauswilke.com/dataviz
- Book: Visualization Analysis & Design, Tamara Munzner, CRC Press, 2014.
- Book: Storytelling with Data, Cole Nussbaumer Knaflic, Wiley, 2015.
- Book: How Charts Lie, Alberto Cairo, W. W. Norton, 2019.
- Article: A Tour Through the Visualization Zoo, Heer, Bostock & Ogievetsky, CACM, 2010. [doi:10.1145/1743546.1743567](https://doi.org/10.1145/1743546.1743567)



Article development led by 
queue.acm.org

**A survey of powerful visualization techniques,
from the obvious to the obscure.**

BY JEFFREY HEER, MICHAEL BOSTOCK, AND VADIM OGIEVETSKY

A Tour Through the Visualization Zoo

THANKS TO ADVANCES in sensing, networking, and data management, our society is producing digital information at an astonishing rate. According to one estimate, in 2010 alone we will generate 1,200 exabytes—60 million times the content of the Library of Congress. Within this deluge of data lies a wealth

Examples

<https://public.tableau.com/app/discover/viz-of-the-day>
<https://community.fabric.microsoft.com/t5/Data-Stories-Gallery/bd-p/DataStoriesGallery>
<https://www.gapminder.org/tools/>
<https://flowingdata.com/>
<https://truth-and-beauty.net/>
<https://www.dataviz-inspiration.com/>
<https://www.data-to-viz.com/>

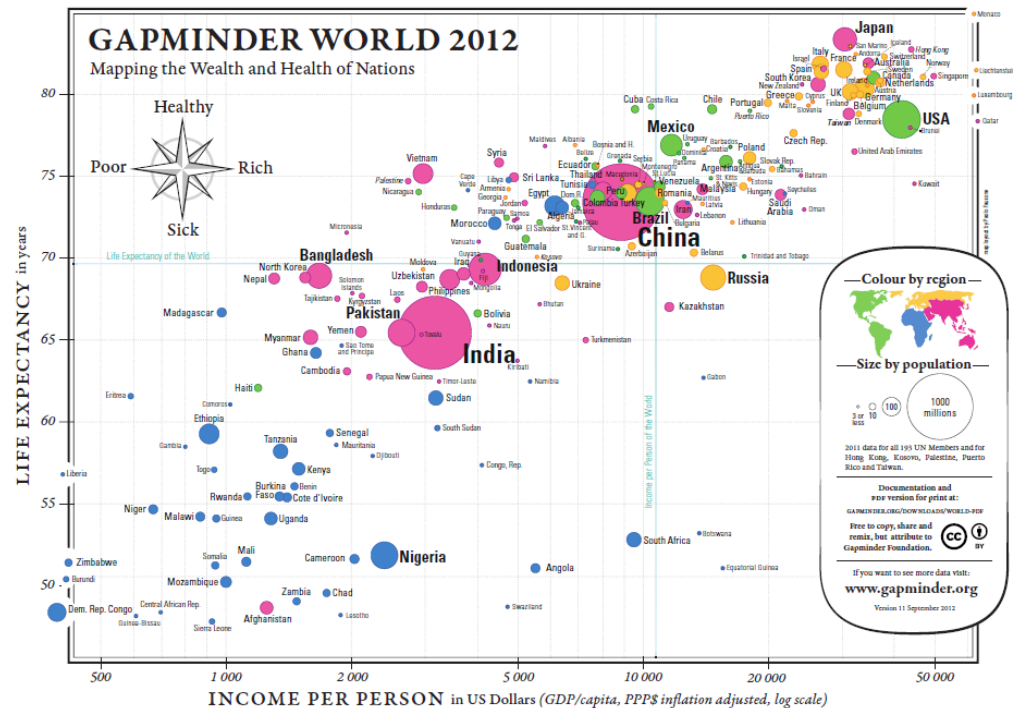
What is Data Visualization?

The process of transforming data
into **visual representations**
that reveal **patterns, relationships, and insights**
to support **understanding, communication, and decision-making.**

Raw data vs. visualization

country	region	income per person	life expectancy	population
Afghanistan	asia_west	1840	57.2	30700000
Albania	europa_east	10400	77	2920000
Algeria	africa_north	13200	76.8	37600000
Andorra	europa_west	41900	82.6	82400
Angola	africa_sub_saharan	6000	61.7	25100000
Antigua and Barbuda	america_north	19100	77	96800
Argentina	america_south	19200	76.1	42100000
Armenia	europa_east	7510	74.3	2880000
Australia	east_asia_pacific	42600	82.3	22800000
Austria	europa_west	44400	80.9	8520000
Azerbaijan	europa_east	15900	70.2	9260000
Bahamas	america_north	23000	73.7	372000
Bahrain	asia_west	41500	76.3	1300000
Bangladesh	asia_west	2710	71.3	156000000
Barbados	america_north	15400	76.8	282000
Belarus	europa_east	17500	71.8	9470000
Belgium	europa_west	41000	80.3	11100000
Belize	america_north	7970	71.6	337000
Benin	africa_sub_saharan	1860	62.6	9730000
Bhutan	asia_west	7030	72.9	753000

Thailand	east_asia_pacific	14400	77.2	67800000
Timor-Leste	east_asia_pacific	2030	72	1160000
Togo	africa_sub_saharan	1260	60.4	6860000
Tonga	east_asia_pacific	5130	70	105000
Trinidad and Tobago	america_north	31300	72.8	1340000
Tunisia	africa_north	10400	77.1	10900000
Turkey	europa_east	20300	78.6	74600000
Turkmenistan	asia_west	12200	68.8	5270000
Uganda	africa_sub_saharan	1640	58.6	36300000
Ukraine	europa_east	8320	71.1	45300000
United Arab Emirates	asia_west	59800	76.4	8900000
United Kingdom	europa_west	36700	80.7	64300000
United States	america_north	50500	78.9	313000000
Uruguay	america_south	18500	76.5	3400000
Uzbekistan	asia_west	4770	69.3	29500000
Vanuatu	east_asia_pacific	2900	63.3	247000
Venezuela	america_south	17700	75.3	29900000
Vietnam	east_asia_pacific	4910	73.6	90500000
Yemen	asia_west	3790	67.9	24900000
Zambia	africa_sub_saharan	3510	54.5	14700000
Zimbabwe	africa_sub_saharan	1850	54.1	14700000



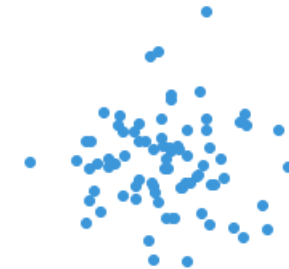
What a simple visualization can do

A scatter plot can reveal
correlational relationships



strong, positive, linear

moderate, negative, linear



null / no relationship

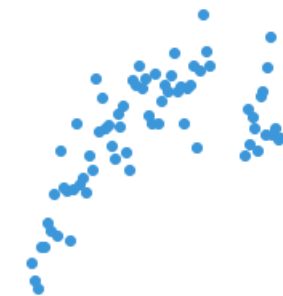


strong, non-linear

and other patterns



data clusters (two)



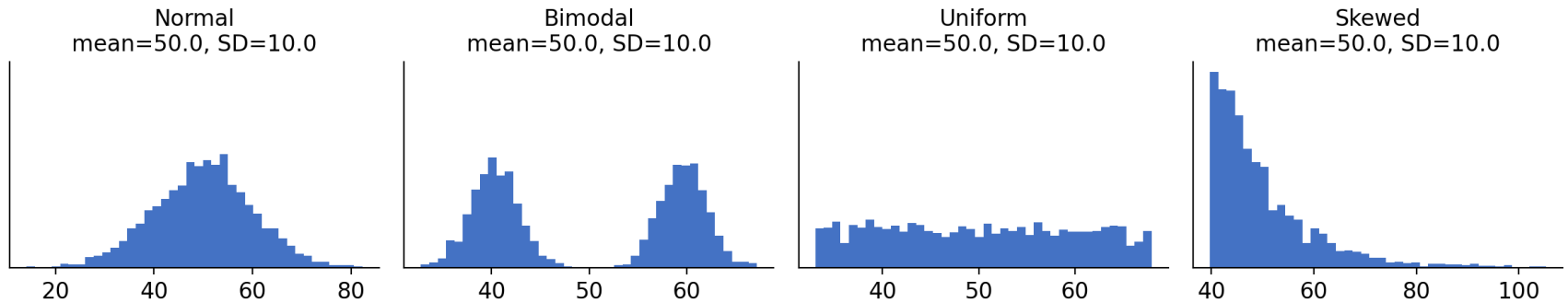
gap in values



outliers (lower left)

Same statistics, different pictures

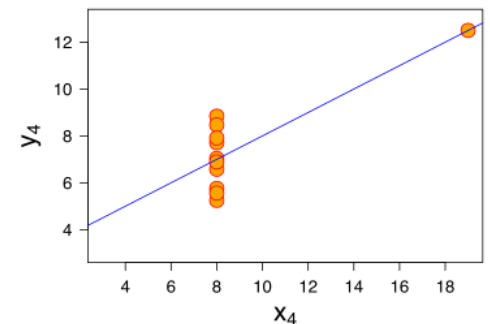
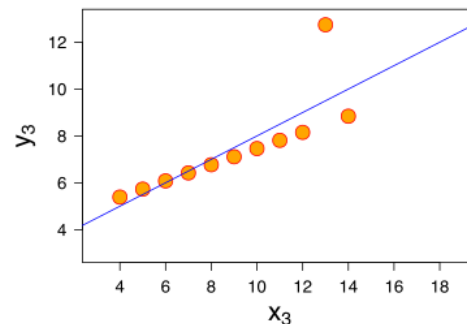
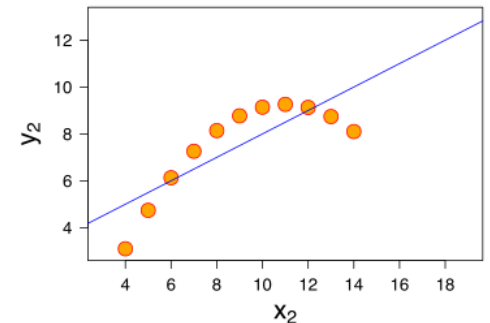
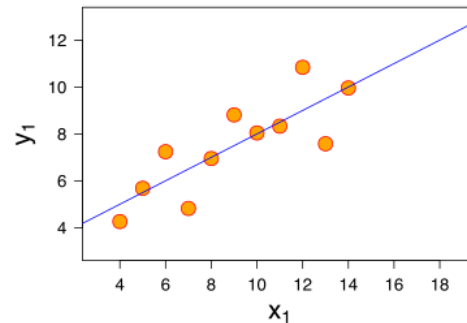
Same mean and standard deviation, different distributions



Anscombe's quartet

- Demonstrate the importance of visual representation of data
- All four datasets have almost identical statistics but different graphs

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89



Property	Value
Mean of x	9
Sample variance of x	11
Mean of y	7.50
Sample variance of y	4.125
Correlation between x and y	0.816
Linear regression line	$y = 3.00 + 0.500x$
Coefficient of determination of the linear regression	0.67

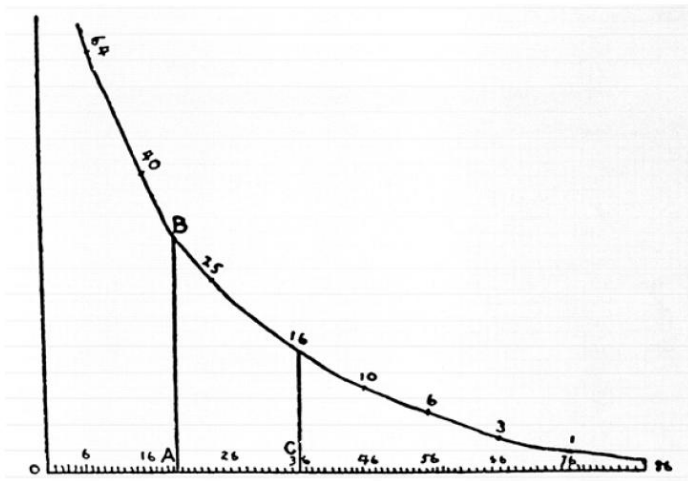
Visualization in the AI era

- AI assistants now write plotting code and draw a chart in seconds.
- What AI cannot decide for you:
 - Which question is worth asking
 - Whether the chart answers it (chart type, encoding, color, scale)
 - Whether the plotted numbers are right
 - What the chart hides, and what it does not prove
- The pattern for every example today: **question → chart choice → implementation → interpretation**
- By the end you can: choose and justify a chart, build it (yourself or with AI), check it, and interpret it cautiously.

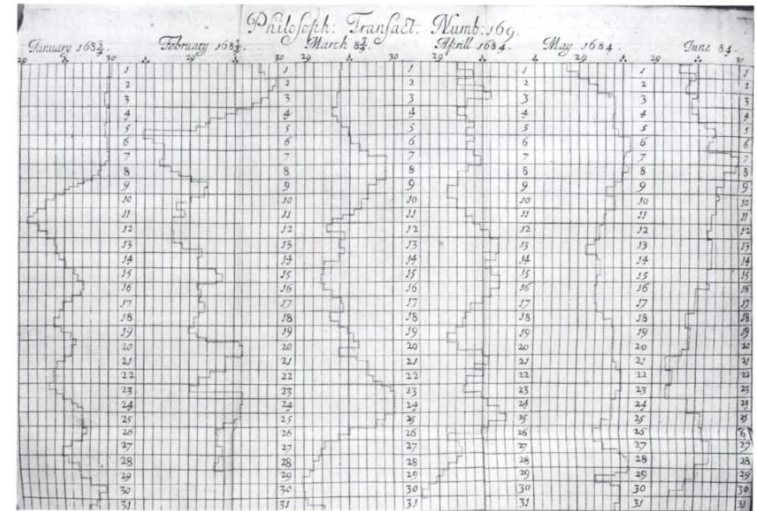
Four types of visualization

- **Relational graphics:** relationships between variables (this lecture)
- **Time series:** change over time (this lecture)
- **Spatial data maps:** data tied to locations (Lecture 2)
- **Network graphs:** relationships between entities (Lecture 3)
- These are broad families, not a strict taxonomy: they can overlap.

Where it began: four early visualizations



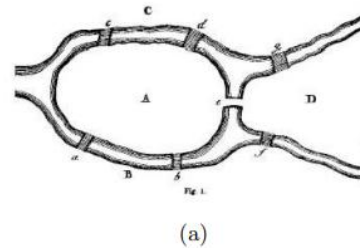
1669 Huygens: survival rate by age
(relational)



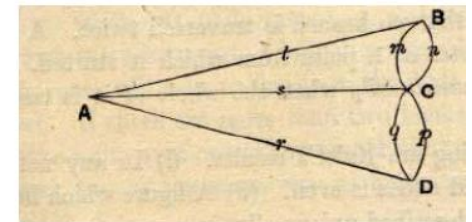
1685 Robert Plot: daily barometric pressure in Oxford
(time series)



1686 Halley: trade winds and monsoons
(map)



(a)

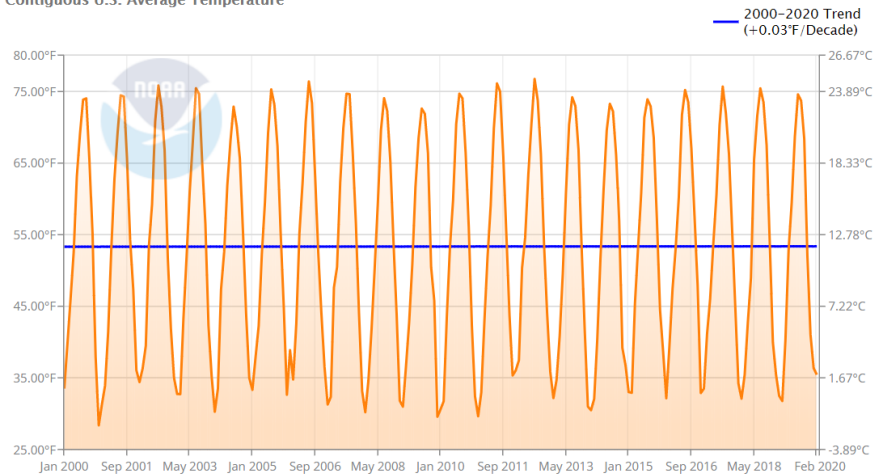


(b)

1736 Euler, 1892 Rouse Ball : the seven bridges of Königsberg
(network)

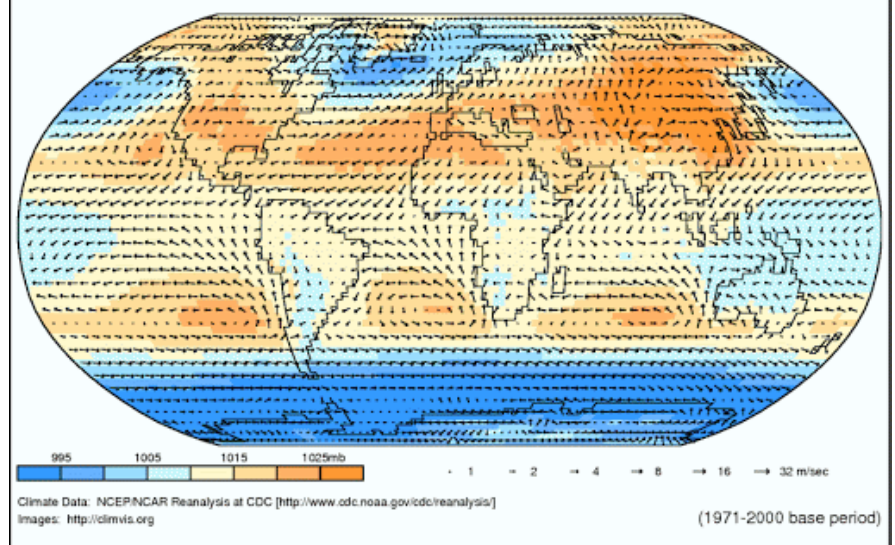
Time series and maps today

Contiguous U.S. Average Temperature



U.S. average temperature, NOAA [Climate at a Glance](https://climate.noaa.gov)

Mean Sea-Level Pressure and Surface Winds



Mean sea-level pressure and surface winds (animated climatology)

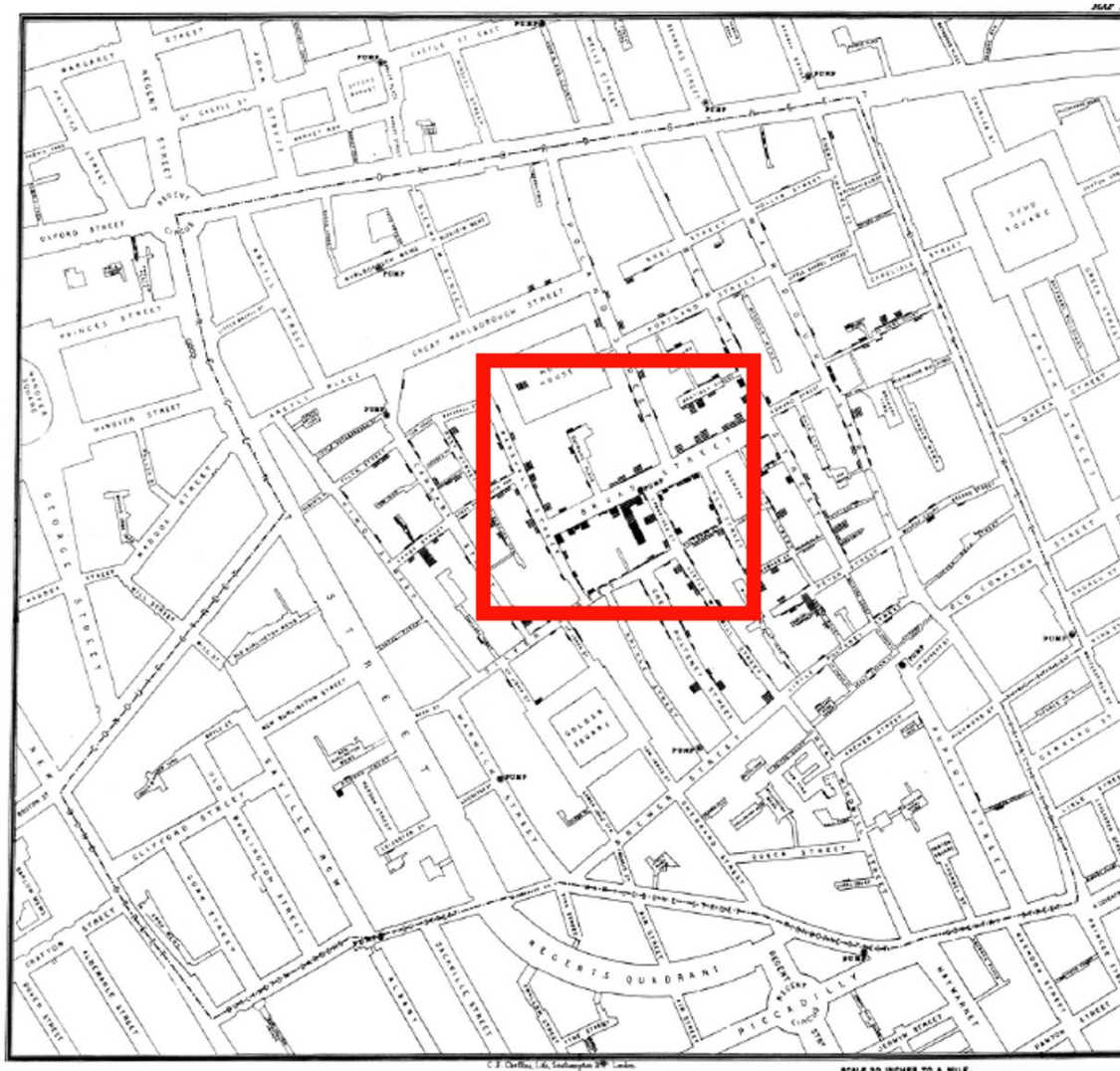
Network Graph

Facebook friendships (2010)

Each line connects between two cities, weighted by the number of friendships and geographic distance.



Classic Example: Dr. John Snow's Cholera Map (1855)



To stop the outbreak of cholera in London in 1854, **Dr. John Snow** marked the **cholera deaths** on a map. This map visualization indicated that the water from a pump on **Broad Street** was to blame as a large number of deaths were marked close to that pump. Snow's visualization is one of the most important early examples of epidemiology, that **clearly linked cholera's spread to water and not air.**

Snow, 1855 in
*On the Mode of
Communication of Cholera*

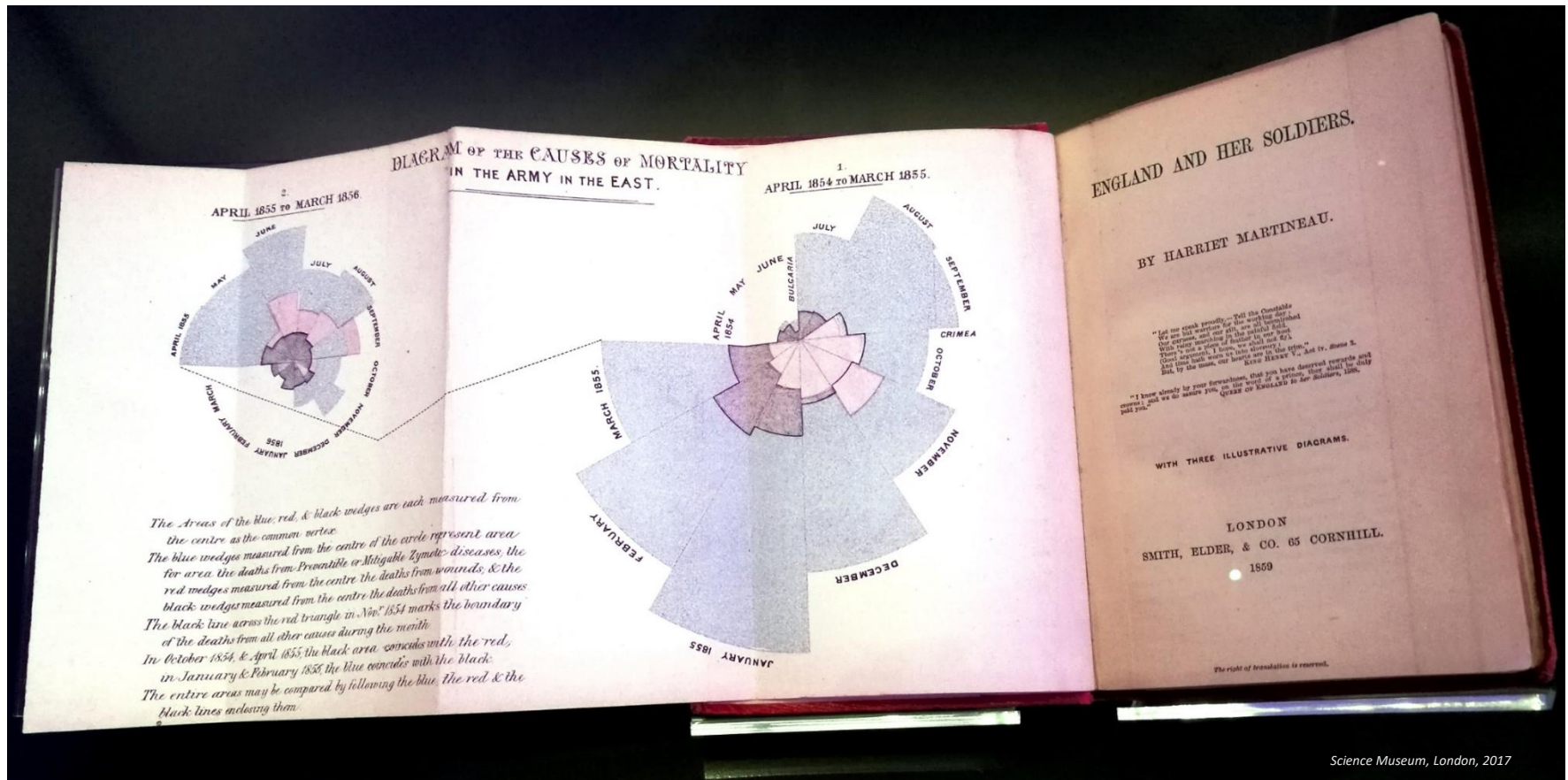
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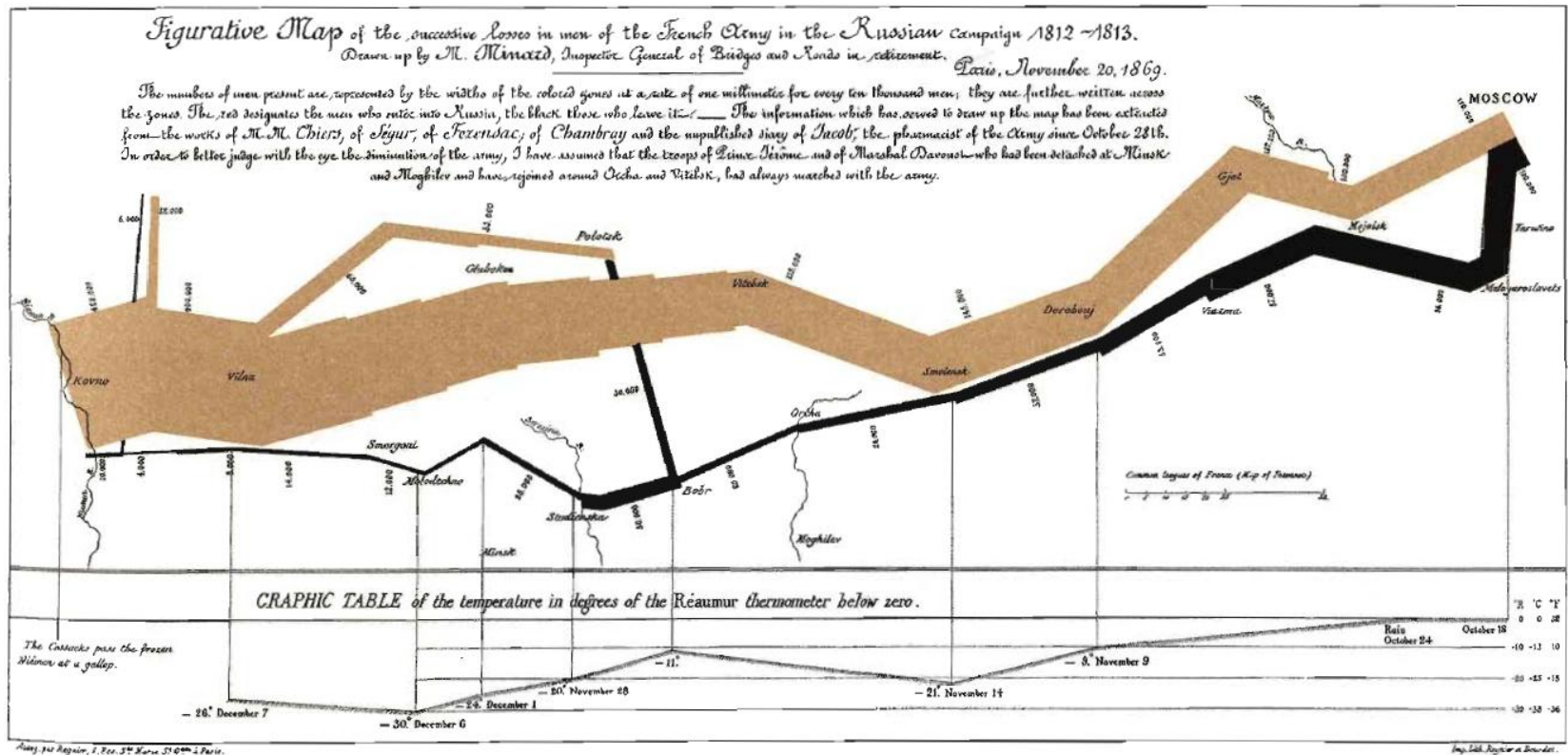
Snow, 1855 in
*On the Mode of
Communication of Cholera*

Classic Example: Florence Nightingale (1859)



- Polar area diagrams (or coxcomb diagram) showed the causes of soldiers' deaths in each month.
- The right diagram showed the data of the first year when she arrived the hospital.
- Blue area showed deaths from preventable diseases due to bad conditions in hospital. Red area showed deaths from battlefield wounds. Black represented other causes.
- The left diagram showed the data of the second year after she implemented hygiene practices.

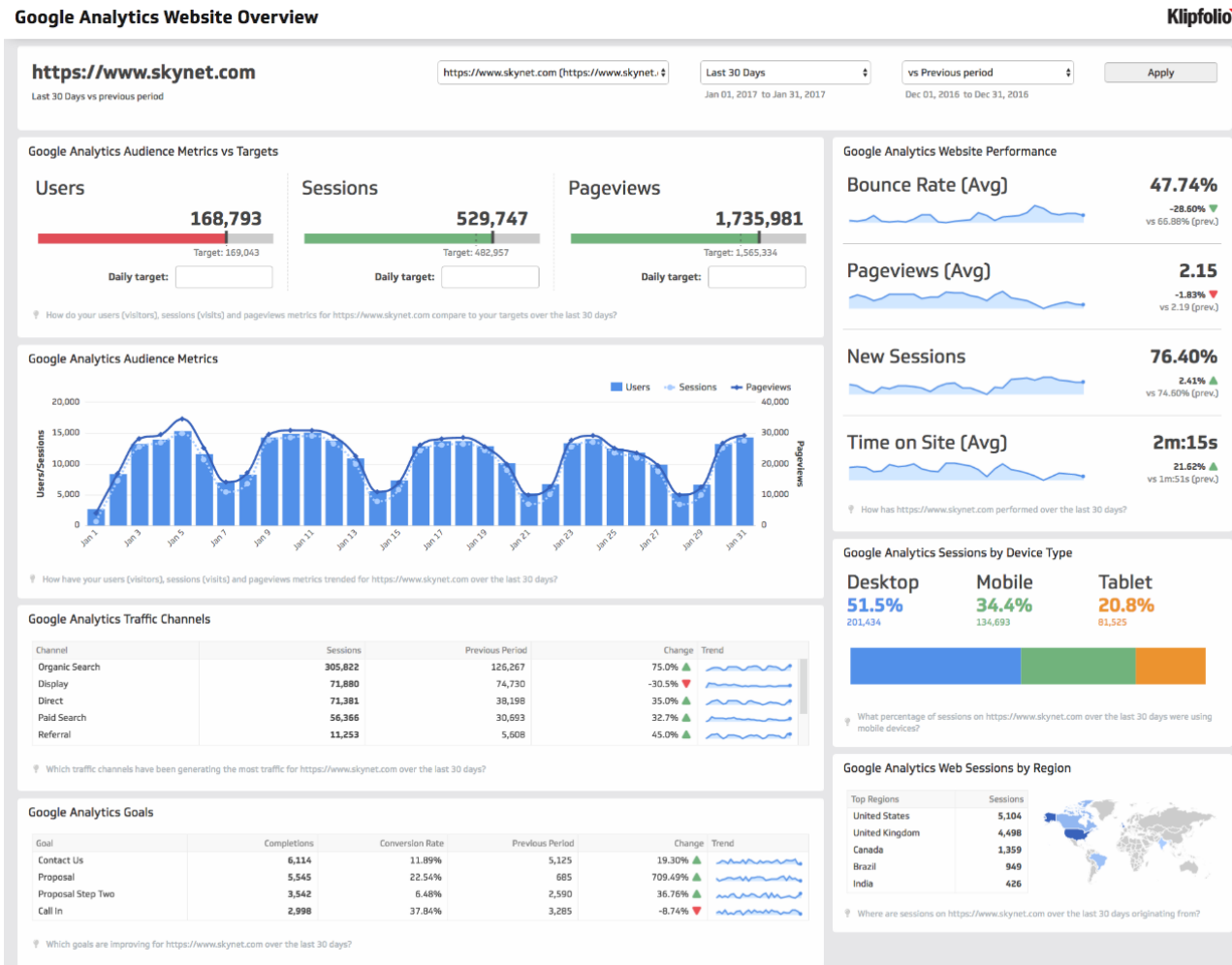
Classic Example: Napoleon's army in Russia (1869)



- Charles Joseph Minard (1869)
- In 1812-1813, Napoleon led the army of 422,000 men from Poland border to Moscow (north-east direction) and then retreated. Only 10,000 men came back.
- Combined data map and time-series
- A flow line represents the route on the map, labeled with place names. Line thickness represents size of army at each place. The different colors represent the directions.
- The time-series chart show temperature and date.
- **Question:** how many variables does this chart show?

Modern Example: Dashboard

Display many measures or key performance indicators (KPIs) on different charts.



Dashboards



Stock market technical chart



JHU COVID-19 dashboard

[Traffic congestion dashboard \(Tableau\)](https://public.tableau.com/app/profile/veera.muangsin/viz/Congestion%20basedonTravelTimeIndex/TravelTimeIndexTTIDashboard)

[https://public.tableau.com/app/profile/veera.muangsin/viz/Congestion basedonTravelTimeIndex/TravelTimeIndexTTIDashboard](https://public.tableau.com/app/profile/veera.muangsin/viz/Congestion%20basedonTravelTimeIndex/TravelTimeIndexTTIDashboard)

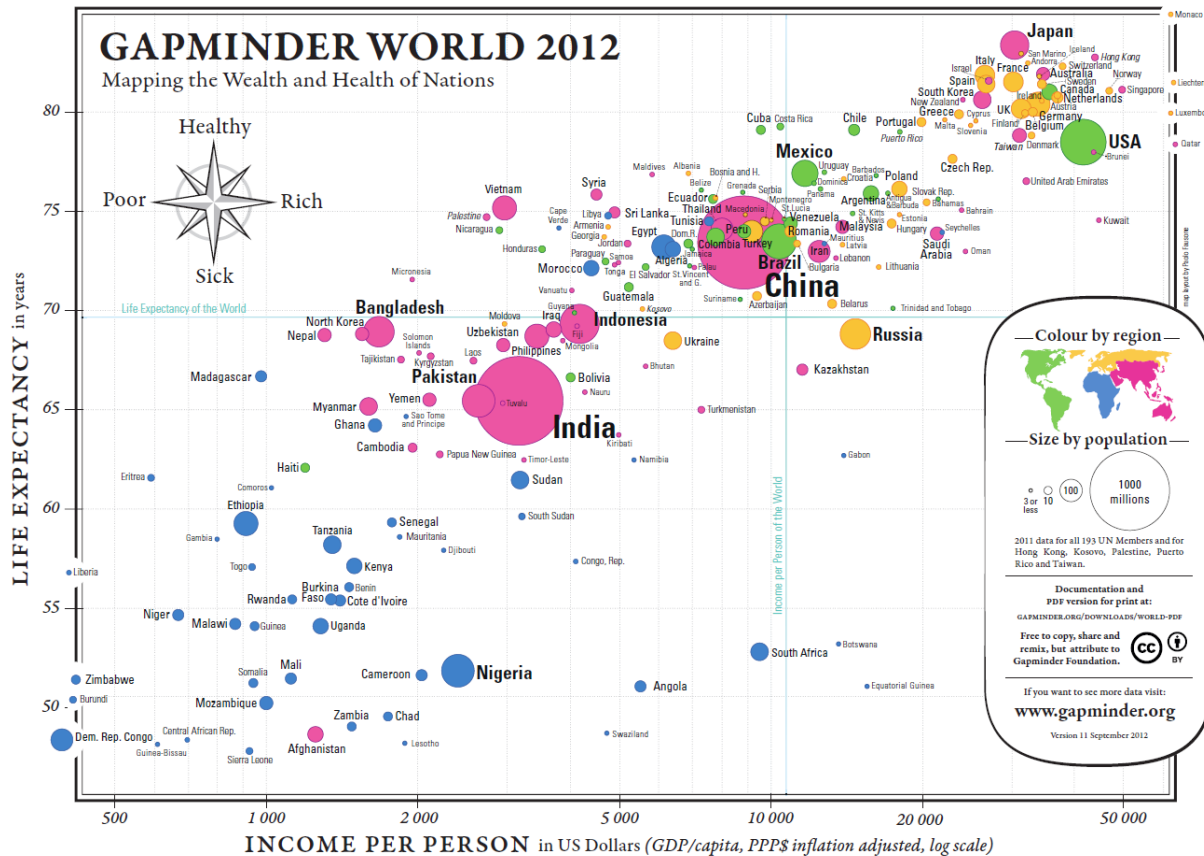
Storytelling: Hans Rosling

“200 Countries, 200 Years, 4 Minutes” (BBC Four, 2010)

[Watch on YouTube](#)

Try it: [Gapminder Tools](#)

Five variables in one animated scatter: income (x), life expectancy (y), population (size), region (color), year (time).



Story Telling with Visualization by Hans Rosling

<https://www.youtube.com/watch?v=QvVIV8jWwzk>

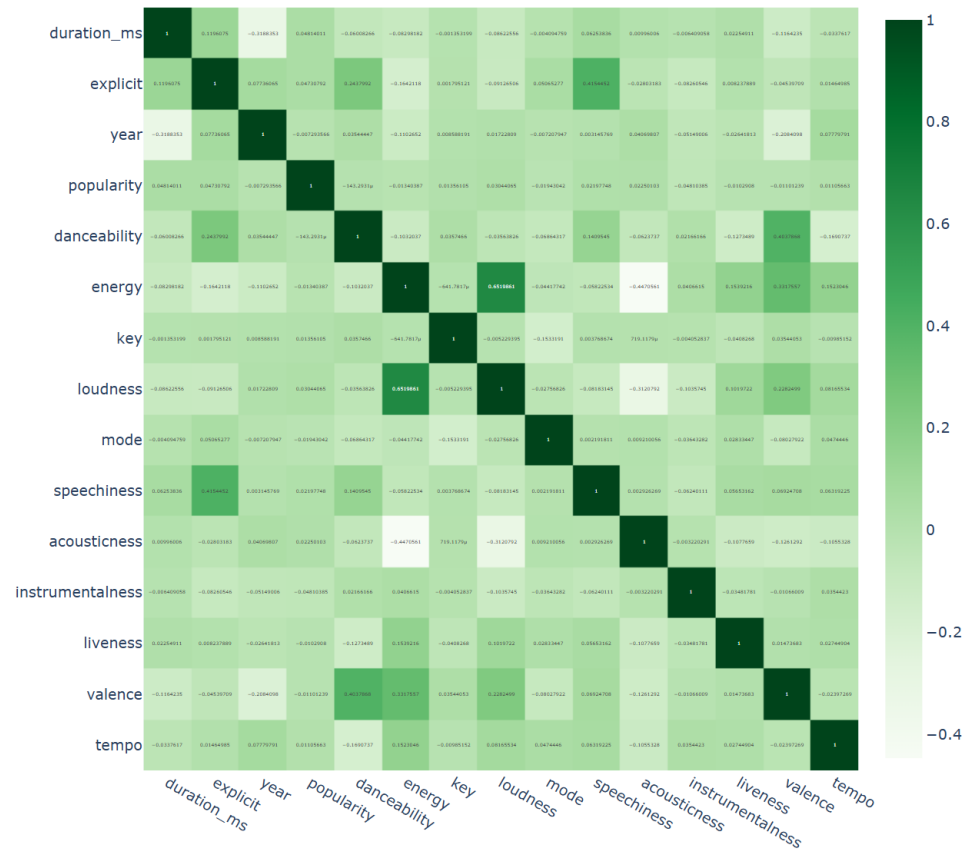
The video is a presentation by Hans Rosling, overlaid on a 3D visualization. The visualization consists of a vertical axis labeled 'life span' with markers for 25 years, 50 years, and 75 years. A horizontal axis is labeled 'income' with markers for \$400, \$4,000, and \$40,000. A large number '1810' is visible in the background. The video player interface includes the YouTube logo, a search bar, and various controls at the bottom.

2.4 - 200 Years, 200 Countries, 4 Minutes

Up next

AUTOPLAY ☐

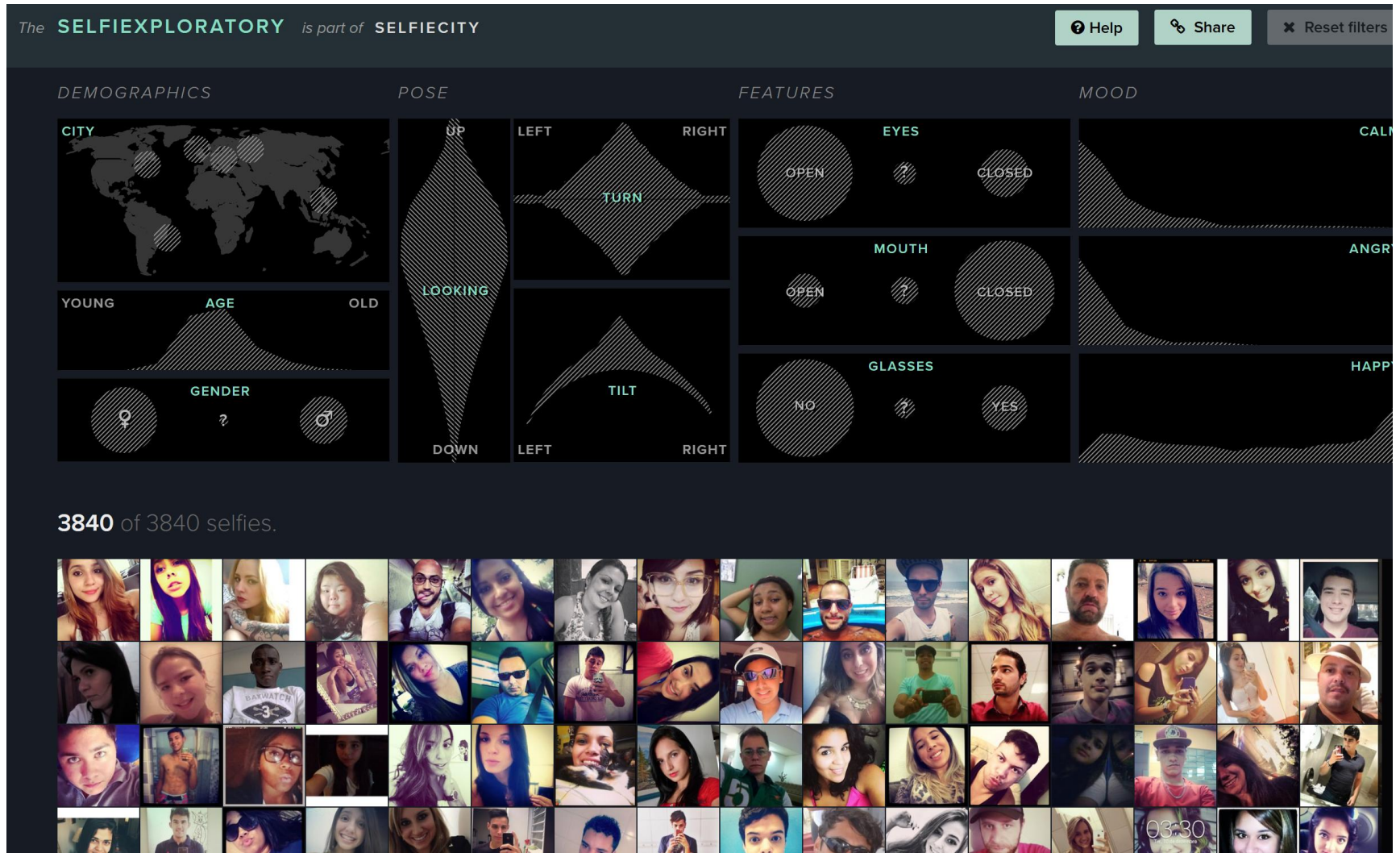
<https://www.kaggle.com/code/varunsaikanuri/spotify-data-visualization>



Example: Visualization in a data science project

SelfieCity

<https://selfiecity.net/> <https://selfiecity.net/selfiexploratory/>



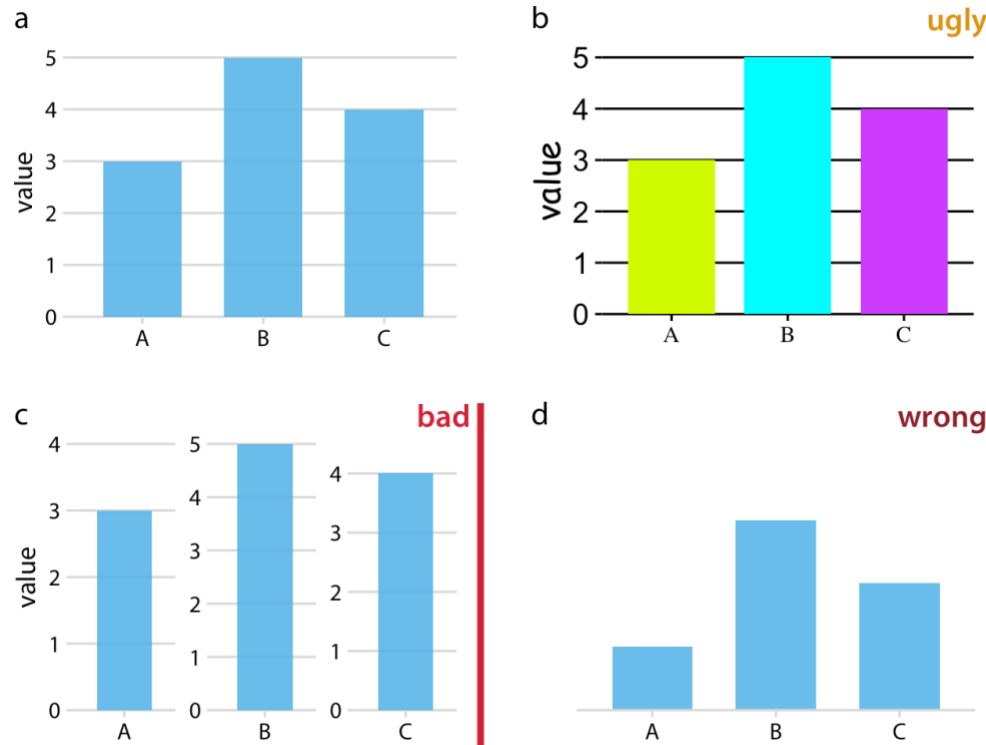
Principles of Effective Data Visualization

What is a good visualization?

- Not just a pretty chart. A good visualization helps the viewer:
 - understand the structure of the data
 - gain insight into the data
 - do tasks more effectively (present, explore, answer questions, decide)
- Is Dr. Snow's map good? Check it against all three.



Ugly, Bad, Wrong



Ugly

A figure that has aesthetic problems but otherwise is clear and informative

Bad

A figure that has problems related to perception; it may be unclear, confusing, overly complicated, or deceiving

Wrong

A figure that has problems related to mathematics; it is objectively incorrect

Graphical excellence: less is more

Edward Tufte: Graphical excellence is well-designed presentation of interesting data,

- consists of **complex ideas** communicated with **clarity**, **precision**, and **efficiency**,
- gives the **greatest number of ideas** in the **shortest time** with the **least ink** in the **smallest space**,
- and **tells the truth** about the data.



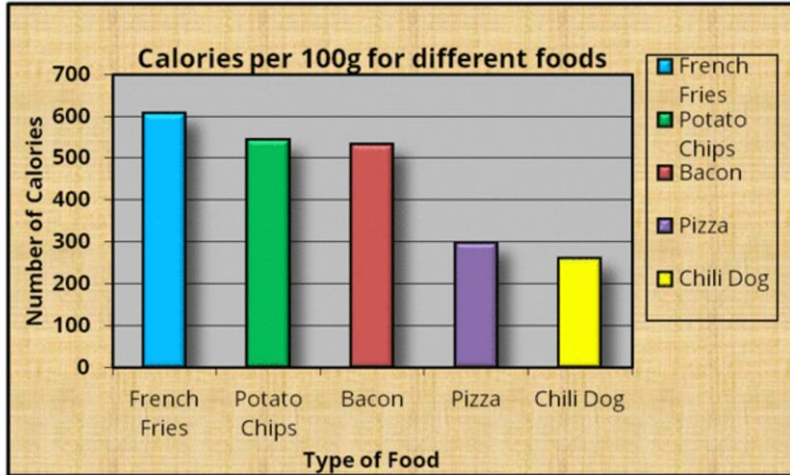
Remove to improve (the data-ink ratio). The author wants to highlight the data for bacon

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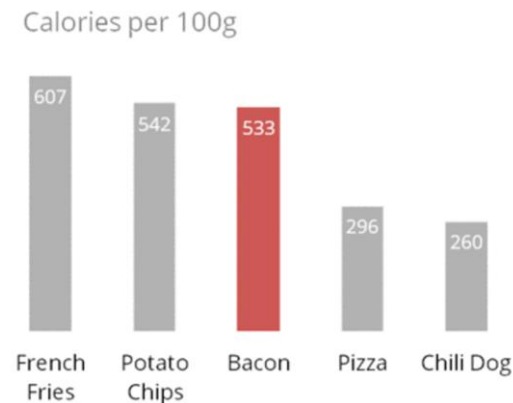
Before



Created by Darkhorse Analytics

www.darkhorseanalytics.com

After

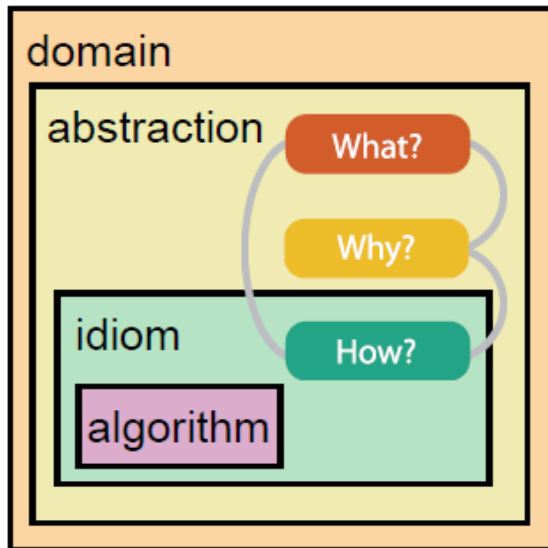


Created by Darkhorse Analytics

www.darkhorseanalytics.com

Remove to improve (the data-ink ratio). The author wants to highlight the data for bacon

Munzner's Visualization Design Methodology



- **What** is shown?
 - Data abstraction
- **Why** is it shown?
 - Task abstraction
- **How** is it shown?
 - Visual encoding

What: know your data

Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box		7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69		5 4-Not Specified	Small Pack	0.44	6/6/05
69		5 4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05

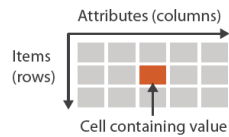
attribute

item

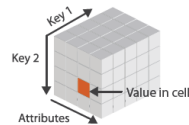
cell

Dataset Types

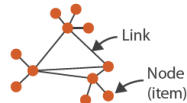
→ Tables



→ Multidimensional Table



→ Networks

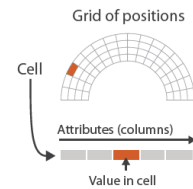


→ Trees



→ Spatial

→ Fields (Continuous)



→ Geometry (Spatial)



Attribute Types

→ Categorical

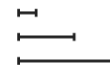


→ Ordered

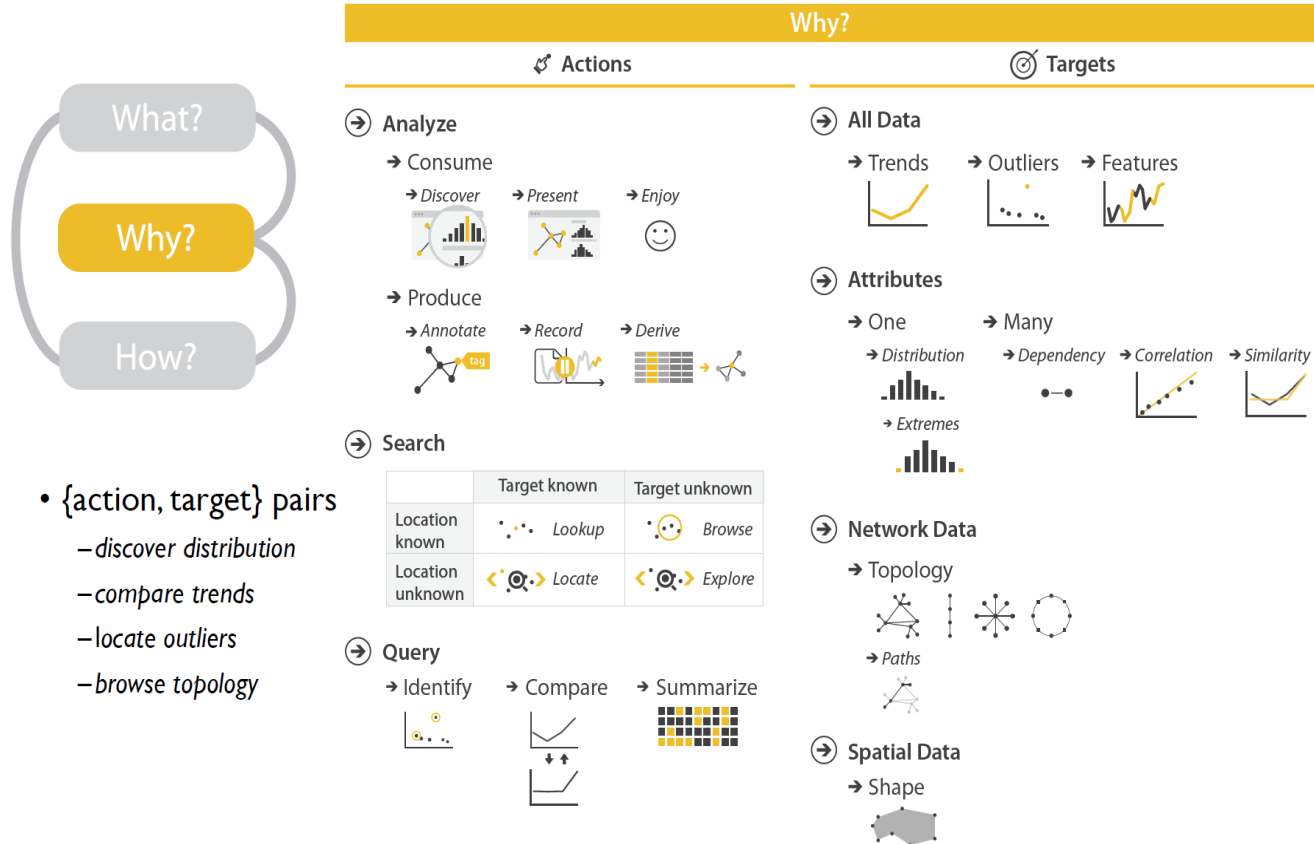
→ Ordinal



→ Quantitative



Why: tasks = {action, target}

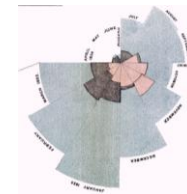
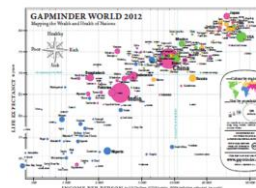


Examples

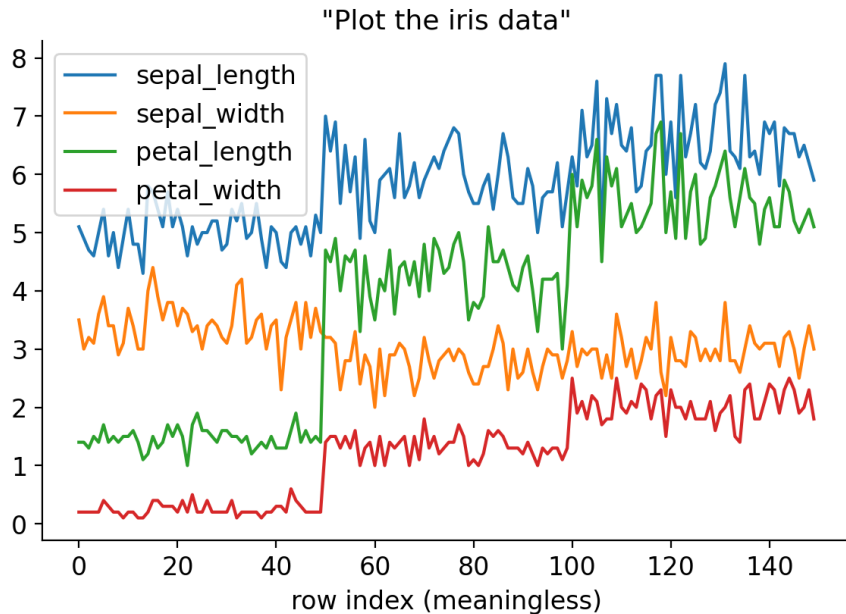
Gapminder: identify a correlation between income and live expectancy

Nightingale: compare distributions by cause of death

Snow: discover the spatial distribution of deaths



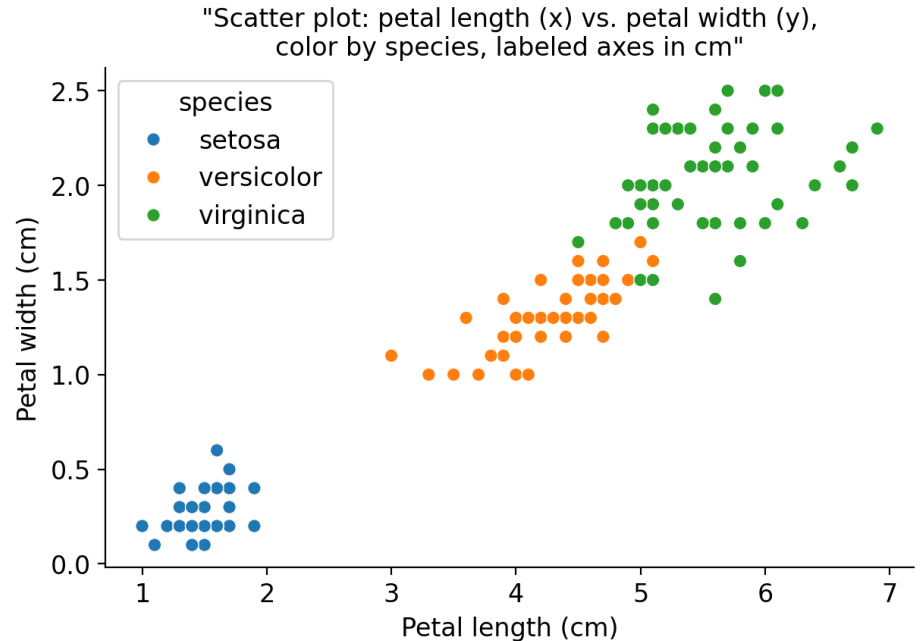
What + Why + How = a good prompt



Vague prompt

"Plot the iris data."

Result: a line chart of all columns against the row index. Correct code, meaningless chart.



What + Why + How

What: 150 flowers, 4 measurements (cm) + species.

Why: compare species by petal size.

How: scatter, petal length on x, petal width on y, color = species, axis labels with units.

Visual Representation

Marks

Geometric primitives used to construct any visualization

→ Points



→ Lines



→ Areas



Visual Variables

The properties or appearance of marks that can be used to represent the data

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Size

→ Length



→ Area



→ Volume



→ Shape

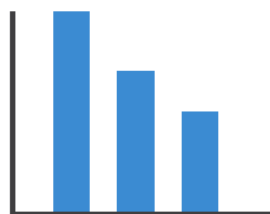


→ Tilt



Visual Encoding

- Combination of marks and visual variables



1:
vertical position

mark: line



2:
vertical position
horizontal position

mark: point



3:
vertical position
horizontal position
color hue

mark: point



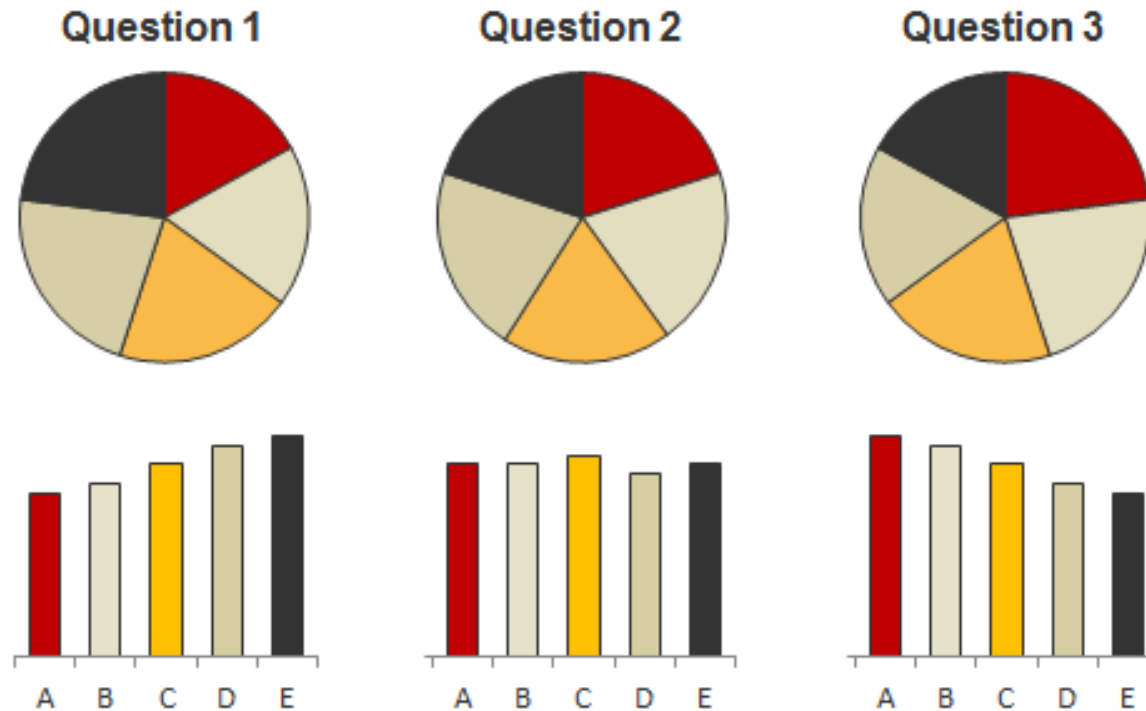
4:
vertical position
horizontal position
color hue
size (area)

mark: point

Effectiveness of Visual Variables

Pie Chart vs Bar Chart

Which one is better? Why?

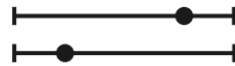


Effectiveness of a visual variable depends on type of data

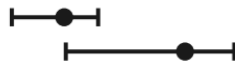
Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



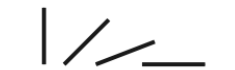
Position on unaligned scale



Length (1D size)



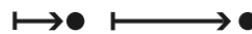
Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



Most

Effectiveness

Least

➔ **Identity Channels: Categorical Attributes**

Spatial region



Color hue



Motion

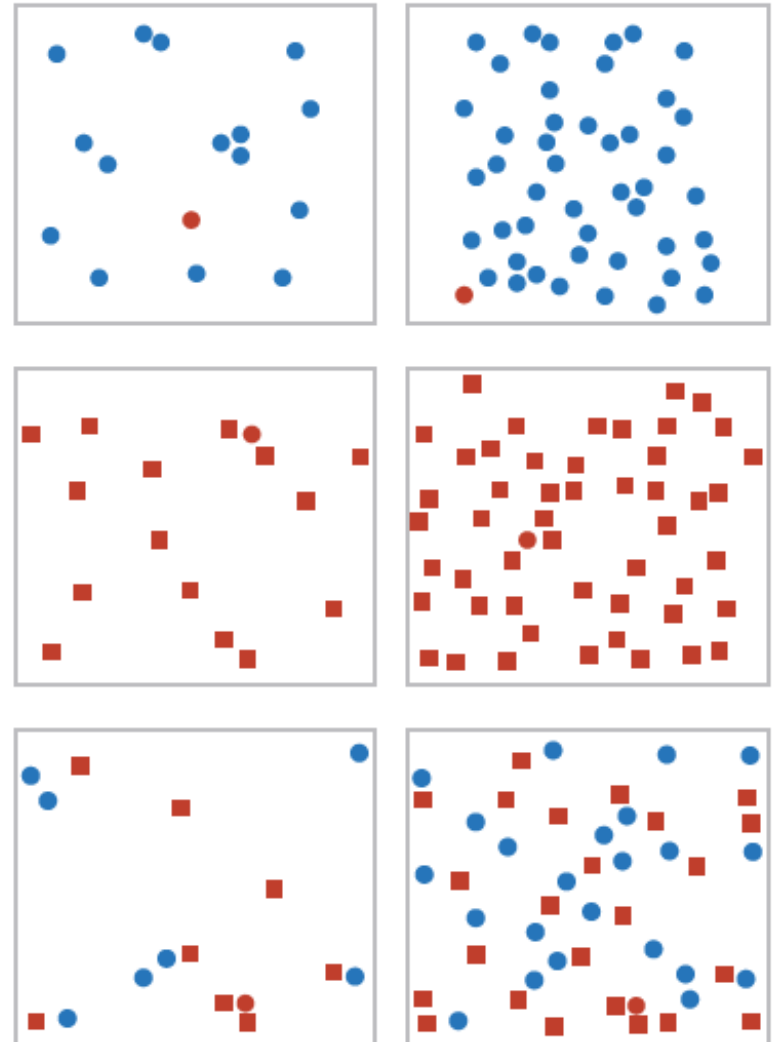


Shape

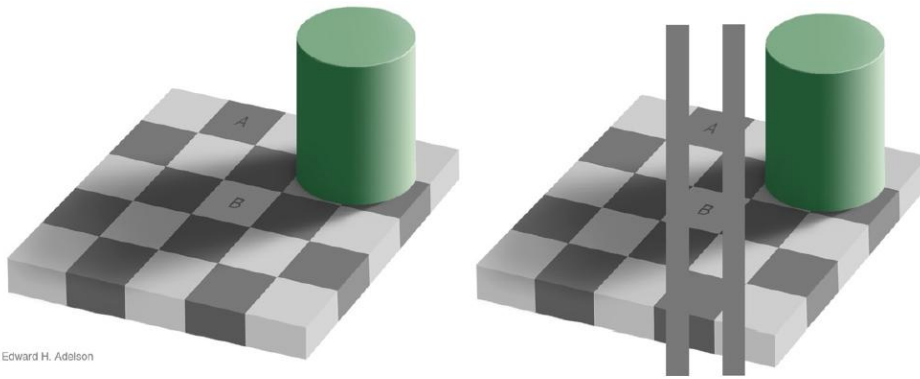


Popout Effect

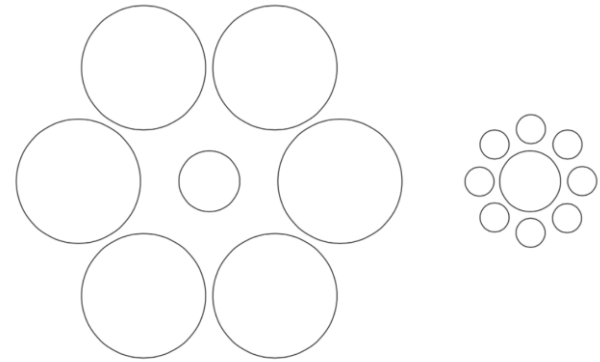
- Find the **red circular dot**
- Popout: a distinct item stands out from many others immediately
- Speed is independent to number of distractors.
- Different visual variables have different level of popout effect.
- E.g. color has stronger popout effect than shape.
- Combination of visual variables weakens popout effect.
- Serial search is needed.
- Speed depends on number of distractor.



Perception can mislead



Luminance is judged against its surroundings



Size is judged against neighbors

The same color or size can read differently in different parts of a chart: compare values on a common scale, not by eye.

Color and order

Luminance



Saturation

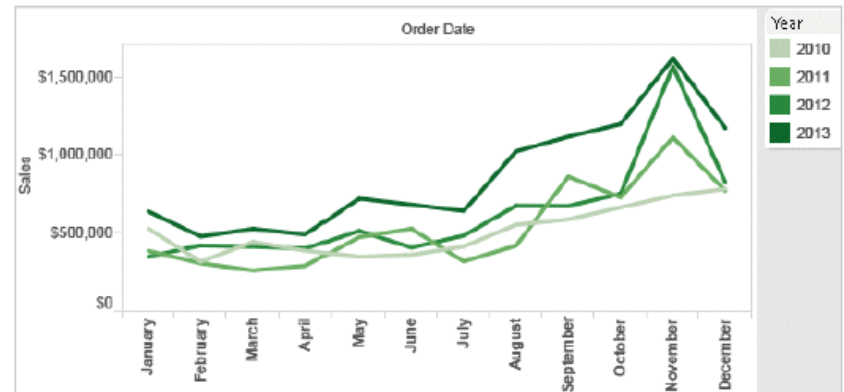
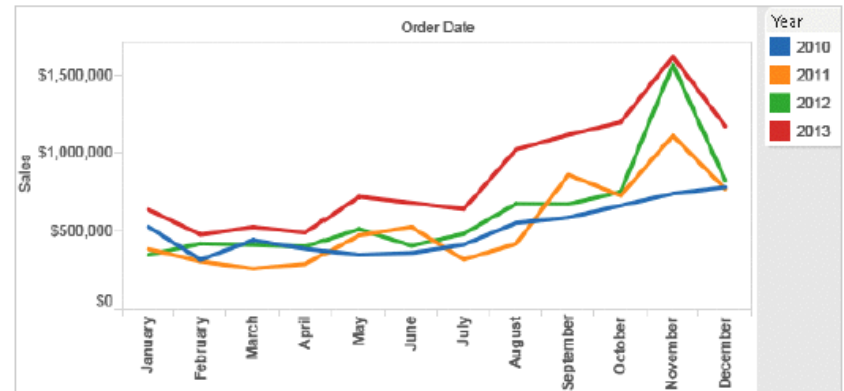


Hue



- **Hue** is unordered: use it for categories
- **Luminance / saturation** are ordered: use them for quantities

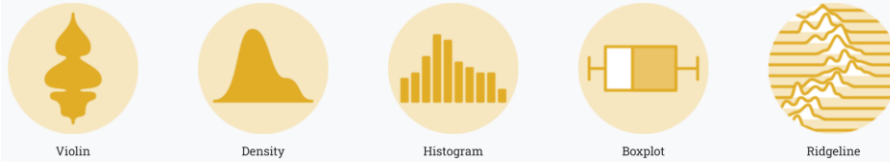
Categorical vs ordered color



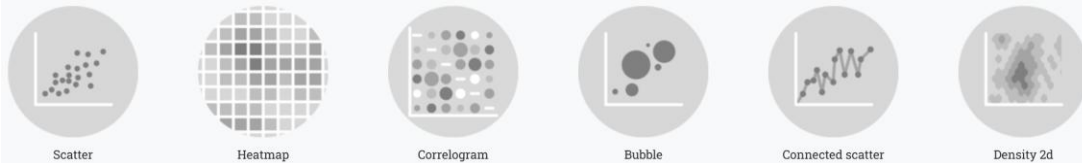
Choosing the Right Chart

Chart Types

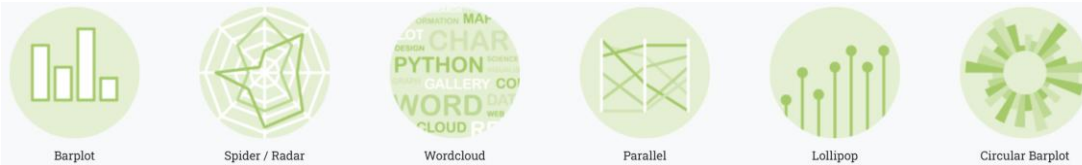
Distribution



Correlation



Ranking



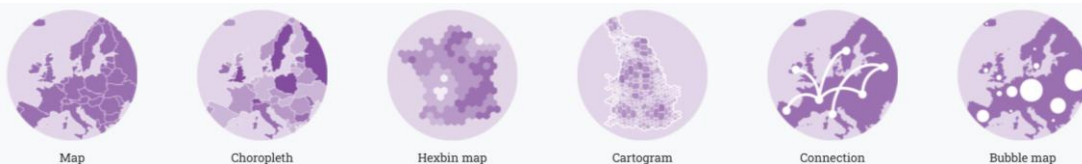
Part of a whole



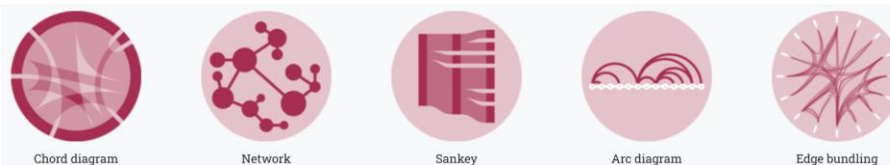
Evolution

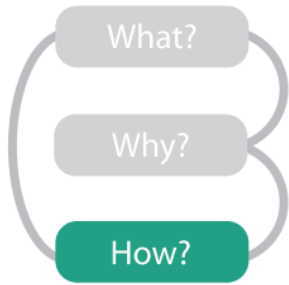


Map



Flow





How to select chart type

Based on

- Data [**What** is shown?]
- Task [**Why** look at it?]
- Effectiveness of visual variables
- Other factors (e.g. scalability)

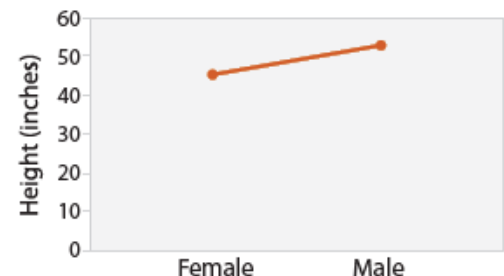
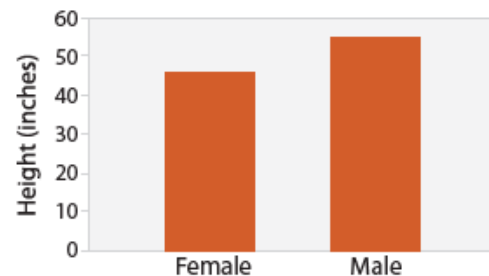
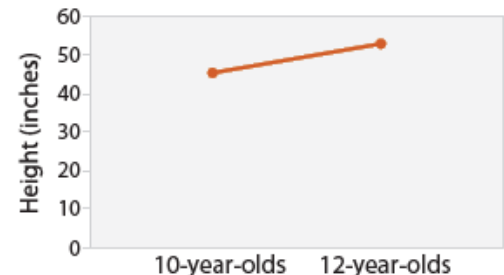
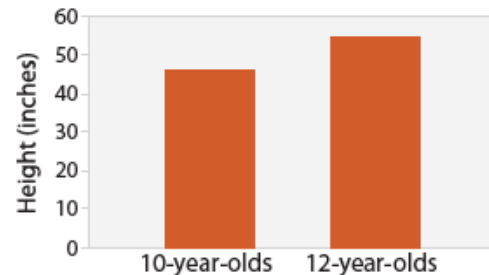
Choosing a chart by task

Task	Chart types	Marks and channels	Common pitfall
Compare values across categories	Bar chart (sorted), dot plot	Length / position on a common scale	Truncated axis; pie with many slices
See the distribution of one variable	Histogram, box plot, violin, density	Position; bar height = count	Bin width changes the story
Show a trend over time	Line chart	Position; slope	Uneven time gaps; dual axes
Show the relationship between two variables	Scatter plot, pair plot	x/y position; color for groups	Overplotting; correlation is not causation
Show parts of a whole	Stacked or 100% bar; pie (few parts)	Length; angle / area	Too many parts; comparing unaligned segments

Box plot: box = Q1 to Q3, line = median. Whiskers reach the most extreme points within $Q1 - 1.5 \cdot IQR$ and $Q3 + 1.5 \cdot IQR$ (Tukey); points beyond are drawn as outliers.

Choosing bar vs line charts

- depends on type of key attrib
 - bar charts if categorical
 - line charts if ordered
- do not use line charts for categorical key attribs
 - violates expressiveness principle
 - implication of trend so strong that it overrides semantics!
 - “The more male a person is, the taller he/she is”



Histogram: different bin sizes tell different stories

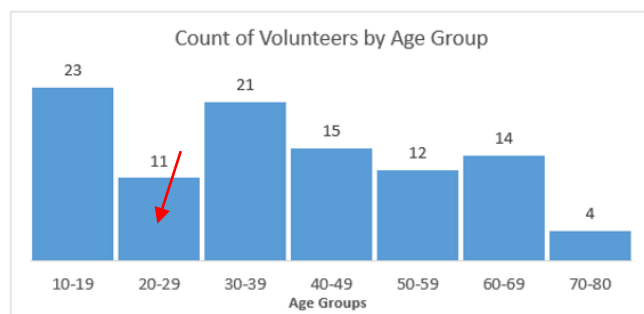
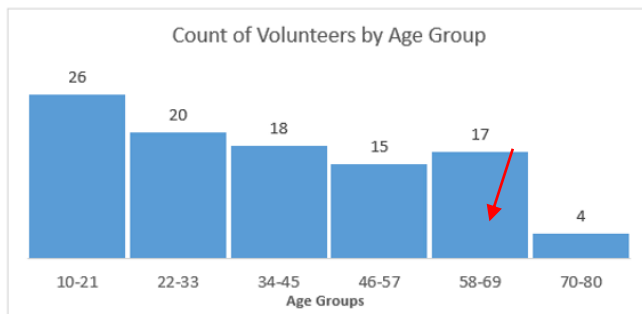
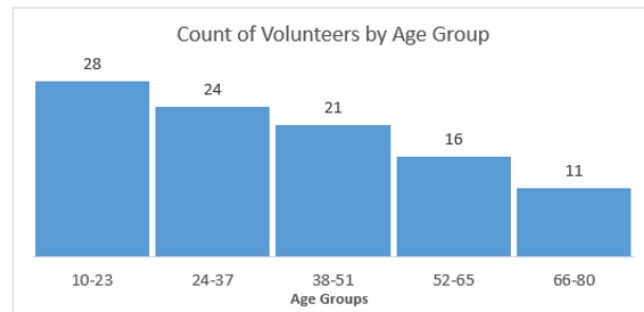
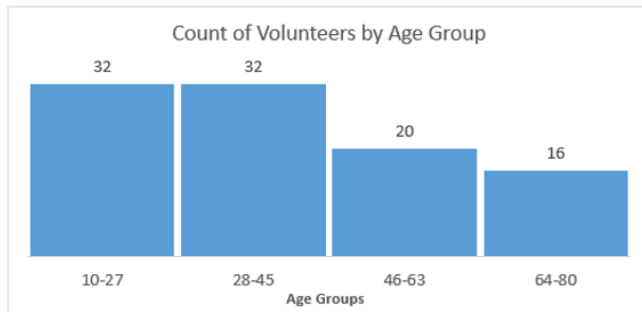
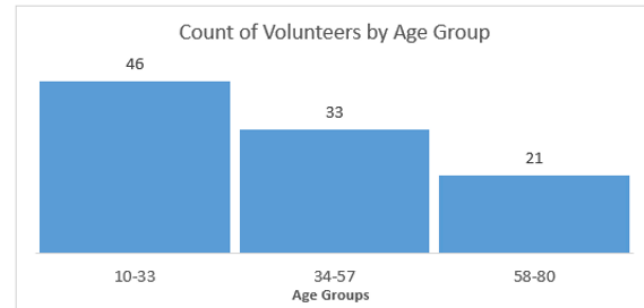
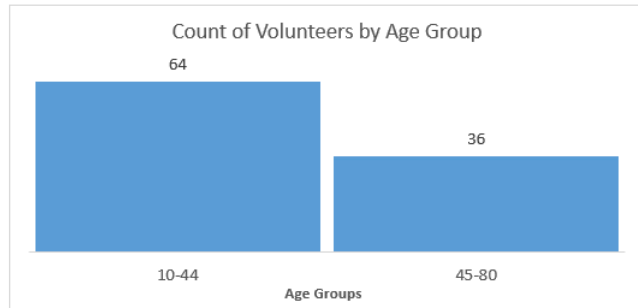
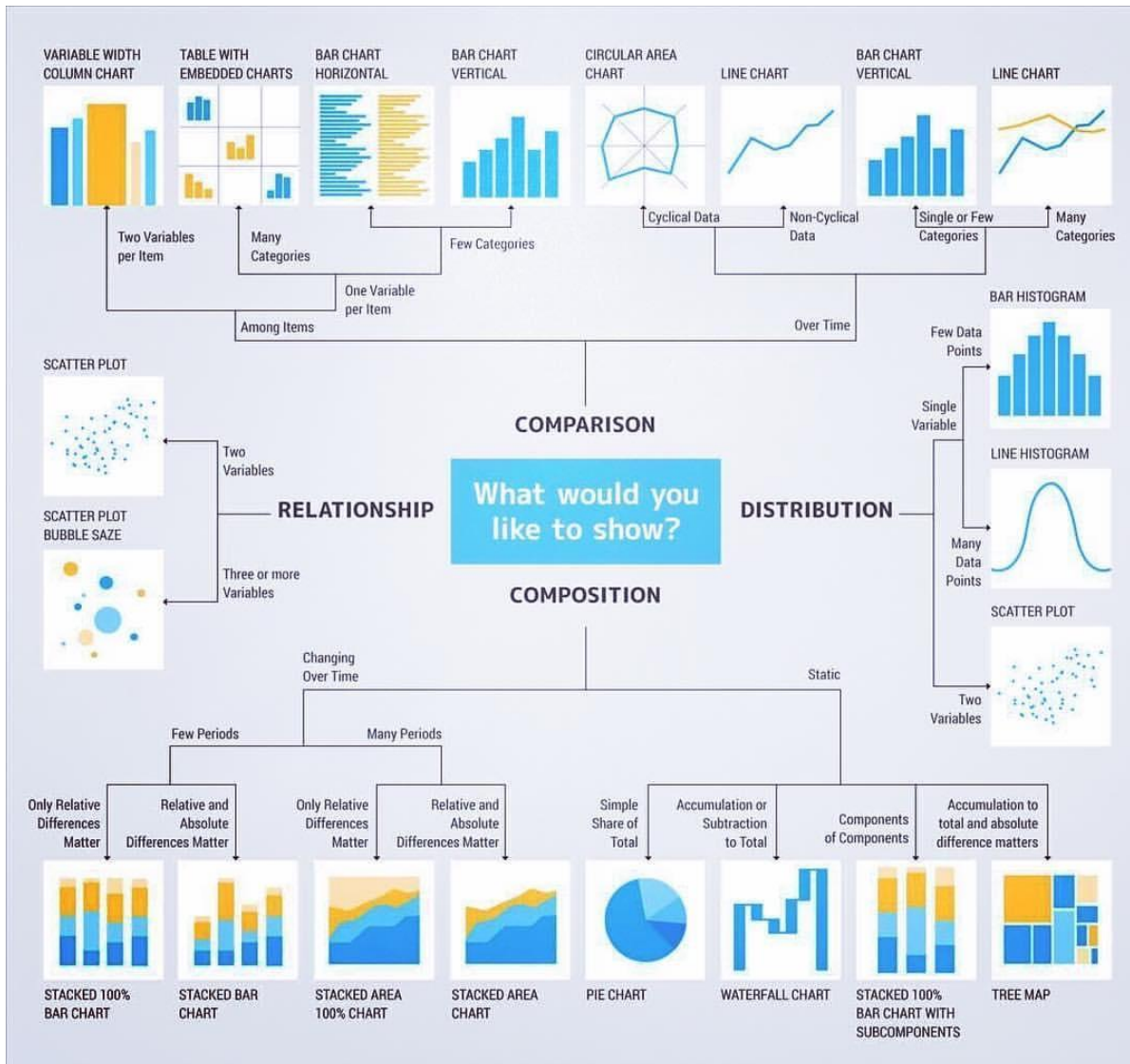


Chart choosers



Start from the data type and the task, then pick a chart.

data-to-viz.com: decision tree from data type to chart, with Python code

Chart chooser (left): what would you like to show? comparison, relationship, distribution, composition

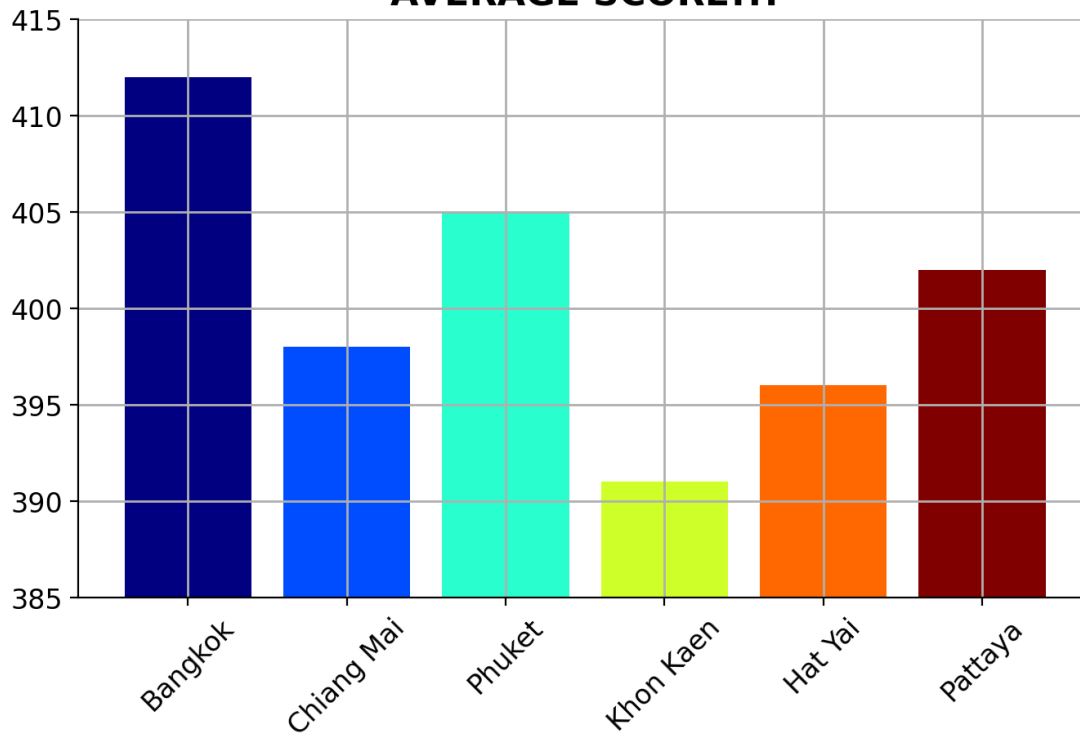
Chart critique checklist

Check	Ask yourself
Question	Does the chart answer one clear question?
Data	Right rows, units, aggregation and denominator?
Encoding	Is the most important variable on position or length?
Color	Right palette type? Color-blind safe? No rainbow?
Axes and scale	Zero baseline for bars? Linear or log? No misleading dual axes?
Ordering	Are categories sorted meaningfully?
Labels and units	Title, axis labels, units, source?
Honesty	No size/volume distortion, no cherry-picked range?
Interpretation	What does it hide? Is a pattern being read as a cause?

Exercise: repair a misleading chart

In pairs, 5 minutes

AVERAGE SCORE!!!



1. List every problem, using the checklist.
2. Sketch a better version, or write the prompt that would produce it.
3. Which problem changes the message most?

Data Visualization Tools

Type	Examples	Specialty
Dashboard	PowerBI, Tableau, Looker Studio (aka Google Data Studio)	
	Grafana	monitoring
Software package	ArcGIS, QGIS	map
	Gephi	network
Programming libraries	Python matplotlib, plotly, streamlit	
	Python streamlit, dash	web app
	Python folium, pydeck	map
	Python networkX, scikit-network, pyvis	network
	Javascript D3	
	R ggplot	